

Relativistic Motion in a Uniform Electric Field

Source: D. Tong, *Electromagnetism*, Problem Sheet 3, Question 9.

A particle of rest mass m and charge q moves in a constant uniform electric field $\mathbf{E} = (E, 0, 0)$. It starts from the origin with initial momentum $\mathbf{p} = (0, p_0, 0)$. Show that the particle traces out a path in the (x, y) plane given by

$$x = \frac{\mathcal{E}_0}{qE} \left(\cosh \left(\frac{qEy}{p_0 c} \right) - 1 \right), \quad (1.1)$$

where $\mathcal{E}_0 = \sqrt{p_0^2 c^2 + m^2 c^4}$ is the initial kinetic energy of the particle.

Solution. We start with the covariant form of the Lorentz force:

$$m \frac{dU^\mu}{d\tau} = q F^{\mu\nu} U_\nu. \quad (1.2)$$

Given $U^\mu = (\gamma c, \mathbf{p})$, it becomes

$$\frac{d\mathbf{p}}{dt} = q\mathbf{E}, \quad (1.3)$$

where $\mathbf{p}$ is the relativistic 3-momentum, namely $\mathbf{p} = \gamma m \dot{\mathbf{x}}$. Thus, we can write

$$\frac{d\mathbf{x}}{dt} = \frac{\mathbf{p}}{\gamma m} = \frac{\mathbf{p}c^2}{\mathcal{E}}. \quad (1.4)$$

This gives

$$\begin{cases} \dot{x} = \frac{p_x c^2}{\sqrt{\mathcal{E}_0^2 + p_x^2 c^2}}, \\ \dot{y} = \frac{p_0 c^2}{\sqrt{\mathcal{E}_0^2 + p_x^2 c^2}}. \end{cases} \quad (1.5)$$

Since we can easily solve for $p_x = qEt$, we can solve the two equations and find

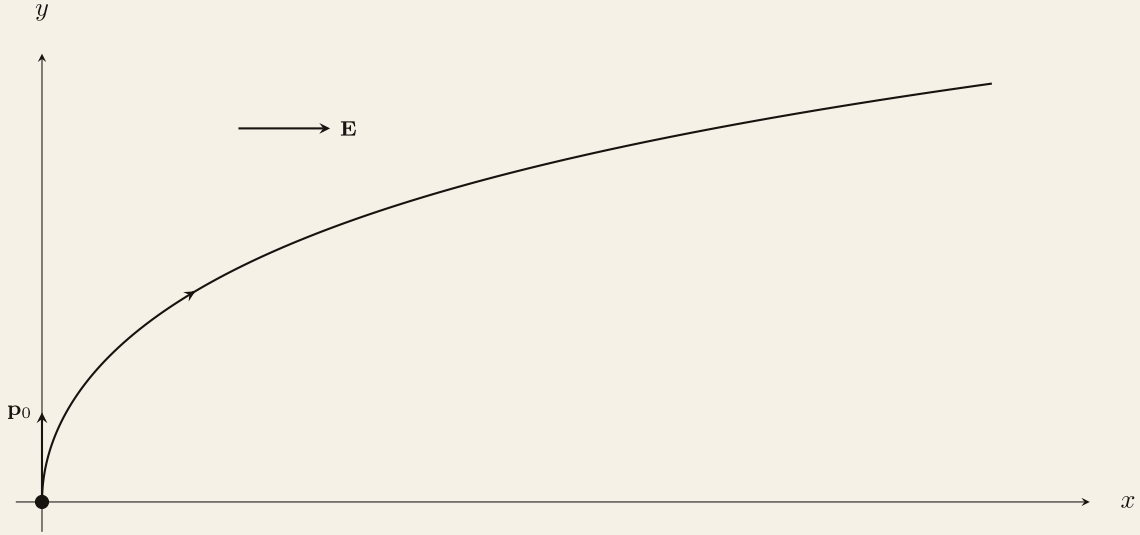
$$y = \frac{p_0 c}{eE} \sinh^{-1} \left(\frac{cqEt}{\mathcal{E}_0} \right), \quad (1.6)$$

and

$$x = \frac{c}{qE} \left(\sqrt{\mathcal{E}_0^2 + q^2 E^2 t^2 c^2} - \mathcal{E}_0 \right). \quad (1.7)$$

Putting these together, we find

$$x = \frac{\mathcal{E}_0}{qE} \left(\cosh \left(\frac{qEy}{p_0 c} \right) - 1 \right). \quad (1.8)$$



Trajectory of a relativistic charged particle in a uniform electric field $\mathbf{E} = (E, 0, 0)$, in natural units $\mathcal{E}_0/qE = p_0 c/qE = 1$. The particle starts at the origin with momentum $\mathbf{p}_0$ in the y -direction and follows $x = (\mathcal{E}_0/qE)(\cosh(qEy/p_0 c) - 1)$.